

Protecting What Matters Most: Hierarchical Digital Transport in Audited Contourlet Steganography

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Abstract

A visually convincing stego image is only part of a successful hidden communication system: the concealed message must also remain identifiable, decodable, and verifiably intact. This study develops an audited hierarchical transport for contourlet-domain image steganography that gives greater protection to the information most responsible for the visual structure of the secret image. The secret is divided into a perceptually dominant Base layer and a complementary Detail layer. Four controlled configurations, C0–C3, separate adaptive coordinate allocation from unequal Reed–Solomon protection while preserving the same 222,360-bit payload and the same realised cover–stego PSNR target of 45.0 ± 0.1 dB. C3 combines transform-aware coordinate ranking with stronger Base-layer coding and Base-first placement in the highest-ranked multiscale coordinate groups. This increases the nominal Base correction budget from 2,752 to 4,160 symbols, a gain of 51.2%, while increasing the Base share of the protected body from 50.0% to 60.2%. The implementation also introduces independently writable P3+P4 range coordinates in an explicit 9–7/PKVA PDFB, a self-describing protected header, layered integrity checks, deterministic coefficient maps, and content-addressed execution evidence. All sixteen clean streams decoded successfully, validating the complete transport path. The tested JPEG, Gaussian, and salt-and-pepper conditions remained beyond the complete-payload recovery region of all four configurations. Together, these results establish a verified Base-layer protection advantage and a clear experimental boundary for the next generation of layered, validity-aware steganographic transport.

Keywords: image steganography; contourlet transform; unequal error protection; adaptive embedding; Reed–Solomon coding; reproducibility.

1 Introduction

Image steganography is often judged first by what an observer can see. When the stego image remains close to its cover, the embedding appears successful. For digital communication, however, visual similarity is only the beginning. The hidden payload must still be identifiable, decodable, and demonstrably intact after compression, noise, quantisation, and the numerical boundaries of the transform implementation.

That distinction shaped this work. We began with a recently reported contourlet-domain pipeline [1]. Reconstructing it required more than translating a sequence of processing labels into code: several choices that directly influence the outcome were not fully specified, including the filter configuration, directional-band organisation, writable-coordinate definition, and datatype transitions around inverse transformation. We made those choices explicit and inspectable, while keeping the source reconstruction separate from the new digital transport developed here.

The reconstruction also exposed a broader opportunity. An embedding rule explains how coefficients may be perturbed, but a usable digital transport must do more. It must describe its own format, organise the message, allocate protection, reproduce the same coefficient map at the receiver, and distinguish a verified secret from an image that merely looks plausible.

DIGITAL_A_D was built around a simple idea: protect the information that matters most to reconstruction. The four most-significant bits of each secret pixel form the Base layer; the four least-significant bits form Detail. Nothing is discarded. Instead, the full secret is organised so that protection can follow information importance.

Two complementary mechanisms put that idea into practice. Adaptive allocation, A, ranks eligible multiscale coordinates using coefficient energy, variance, magnitude entropy, and stability measured on separate calibration images. Unequal protection, D, redirects Reed–Solomon redundancy toward Base. C3 combines both mechanisms, giving Base stronger coding and earlier access to the coordinate groups favoured by calibration.

The four configurations make each design choice visible. C0 uses fixed allocation and symmetric coding; C1 adds adaptive allocation; C2 adds unequal Base/Detail protection; and C3 brings the two mechanisms together with Base-first placement. Payload size and realised image quality remain fixed across the comparison, so an apparent advantage cannot be purchased by hiding less information or accepting more visible distortion.

A second contribution arose from the transform itself. The number of entries in a directional representation is not automatically the number of independently writable degrees of freedom. We therefore constructed and verified an explicit multiscale coordinate space over the valid P3+P4 range of a 9–7/PKVA PDFB implementation. The resulting pool is large enough for the complete protected payload and preserves deterministic analysis, embedding, reconstruction, and extraction maps.

The completed experiment produced sixteen embeddings and eighty-eight scheduled evaluations. Every clean stream decoded successfully, confirming that payload construction, coding, coordinate selection, PDFB embedding, inverse transformation, uint8 quantisation, semi-blind extraction, and integrity checking work together as one transport. C3 also delivered the intended structural gain for Base. The selected JPEG, Gaussian, and salt-and-pepper conditions then located an operating region beyond complete-payload recovery for all four configurations. Rather than obscuring the gain, this boundary shows precisely what was strengthened and where further margin is still needed.

2 Study Contributions

This work makes six connected contributions.

1. **Perceptually informed payload organisation.** The secret is divided into Base and Detail according to the contribution of each bit plane to the reconstructed pixel value.
2. **Hierarchical Base-layer protection.** The fixed Reed–Solomon budget is redistributed toward Base without reducing the payload or expanding coefficient capacity.
3. **Joint coding and placement prioritisation.** Base receives both stronger codewords and earlier access to higher-ranked multiscale coordinates.
4. **Transform-bound adaptive allocation.** Eligible coordinate groups are ranked using energy, variance, magnitude entropy, and calibration-derived stability tied to the active transform fingerprint.
5. **An independently writable PDFB coordinate space.** Sufficient independent P3+P4 range coordinates are constructed and verified within the explicit 9–7/PKVA interpretation.
6. **Validity-aware, traceable transport.** A self-describing protected header, layered integrity checks, deterministic maps, content-addressed objects, and restart-safe execution connect each claim to inspectable evidence.

The novelty lies in this integration rather than in any one component alone. Semantic layering, unequal coding, transform-aware placement, explicit PDFB coordinates, fixed comparison controls, and verifiable recovery are brought together in one end-to-end system whose operating boundary can be examined rather than inferred.

3 Related Work and Novelty Positioning

Transform-domain steganography uses image structure to balance embedding capacity, visual quality, and payload recovery [2, 8, 9]. Contourlet representations are attractive because their multiscale and directional components describe edges and image activity more flexibly than a single-scale transform.

Table 1: Closest methodological lines and the integrated opportunity addressed in this study.

Literature line	Established capability	Integrated advance developed here
Source contourlet study [1]	LPDFB embedding combined with AP/GP/HP processing and attack experiments	Explicit transform interpretation plus a separate, self-describing C0–C3 digital transport
Adaptive contourlet methods [3, 4]	Cover or region selection and content-aware embedding in contourlet representations	Transform-bound stability, fixed-payload comparison, and joint Base coding and coordinate priority
UEP steganography [5]	Unequal channel protection for embedded image information	Semantic Base/Detail hierarchy integrated with multiscale coordinate ranking and common realised PSNR
Adaptive JPEG robustness [7]	Joint selection of embedding domain and adaptive error correction for lossy JPEG channels	Contourlet-bound calibration, independently writable PDFB coordinates, and validity-aware C0–C3 decomposition
RS-based image communication [6]	Layered Reed–Solomon protection for image transmission	Protected metadata, deterministic coefficient mapping, complete integrity validation, and content-addressed evidence

Earlier contourlet studies demonstrated content-aware cover selection, region selection, and adaptive embedding [3, 4]. Unequal error protection has also been applied to embedded image information and layered image communication [5, 6], while more recent work combines embedding-domain selection with adaptive error correction for lossy JPEG channels [7]. The present contribution does not claim any of these ingredients individually. It connects them through semantic Base/Detail organisation, a fixed redundancy budget, transform-bound placement, independently writable PDFB coordinates, common payload and quality controls, and self-describing validity.

Table 1 positions this integration relative to the closest methodological lines.

The DIGITAL_A_D path remains separate from the source AP/GP/HP reconstruction. Preserving that separation keeps the new mechanism identifiable and lets its design choices, evidence, and scope be inspected directly.

4 System Architecture

The system is organised around one guiding principle: information should be protected according to the role it plays in reconstructing the hidden image. A 512×512 grayscale cover carries a 128×128 grayscale secret. Inputs enter the digital path as uint8 images, and clipping plus half-up rounding define the final output boundary. Quality is therefore measured on the image that is actually stored and transmitted rather than on an intermediate floating-point array.

The four most-significant bits of each secret pixel form Base, while the four least-significant bits form Detail. Each layer contains 8,192 raw bytes. The split preserves the complete secret while giving the transport a simple notion of importance: Base carries the coarse intensity structure, and Detail refines it.

Figure 1 shows how three information streams meet at the embedding plan. The cover supplies the multiscale transform coordinates. The secret supplies the independently coded Base and Detail streams. Separate calibration covers supply evidence about coordinate-group stability. Calibration does not use the final test pairs, and the three streams remain separate until the deterministic embedding plan is formed.

At the receiver, the original cover supports semi-blind coefficient differencing. Reed–Solomon and CRC checks then determine whether the extracted stream represents a verified secret. This design connects perceptual organisation, transform placement, and digital validity within one end-to-end path.

4.1 Hierarchical unequal protection

C0 and C1 protect Base and Detail symmetrically. Each layer receives 32 RS(255,127) and 22 RS(255,191) codewords. C2 and C3 retain the same total transport budget but distribute it according to layer importance: Base receives 65 RS(255,127) blocks and Detail receives 43 RS(255,191) blocks.

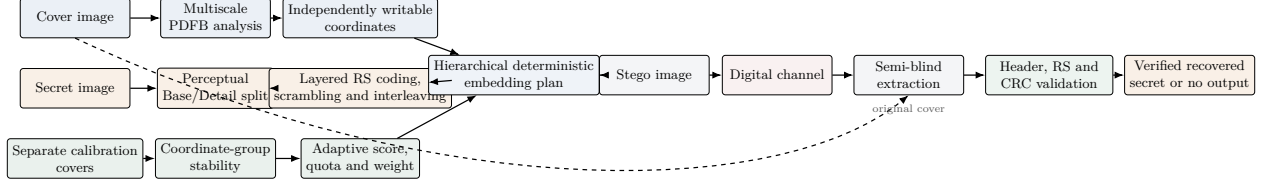


Figure 1: Hierarchical digital transport in an explicit multiscale PDFB coordinate space. The cover defines the writable representation, the secret defines the layered payload, and separate calibration data guide adaptive placement. Recovery is accepted only after the required coding and integrity checks succeed.

The nominal Base correction budgets are

$$\begin{aligned} C0 : \quad & 32(64) + 22(32) = 2,752, \\ C3 : \quad & 65(64) = 4,160 \end{aligned} \quad \text{symbols.} \quad (1)$$

The increase is 51.2%, subject to the per-codeword condition that no block exceeds its own correction radius. Twenty-two Base codewords move from a 32-symbol to a 64-symbol correction limit. Unit tests verify the implemented boundaries by correcting exactly 64 versus rejecting 65 altered symbols for RS(255,127), and correcting 32 versus rejecting 33 for RS(255,191).

This improvement does not create extra channel capacity or reduce the secret. It uses the existing redundancy more deliberately by investing a larger share in the bits that dominate the reconstructed pixel value.

4.2 Transform-bound adaptive allocation

Adaptive methods score each eligible coordinate group using four complementary signals: coefficient energy, variance, entropy of absolute magnitudes, and channel stability measured on designated calibration covers. Energy and variance describe local activity, entropy captures distributional richness, and calibration-derived stability adds evidence about behaviour under representative channel distortions.

Median/MAD normalisation limits the influence of isolated extreme groups. Degenerate feature cases are handled deterministically. The stability artefact records both the calibration attacks and the transform fingerprint, and the final runner applies it only when that fingerprint matches the active transform.

Scores control two connected decisions. They determine integer coordinate quotas and bounded local embedding weights between 0.75 and 1.25. C1 introduces adaptive quota and power while retaining symmetric coding. C2 introduces hierarchical coding while retaining uniform allocation. C3 integrates adaptive quota, adaptive power, unequal protection, and Base-first placement in descending score order.

The integrated design therefore gives Base two complementary advantages: stronger channel coding and earlier access to the coordinate groups favoured by the calibration policy.

4.3 Independent multiscale coordinate capacity

The final profile uses 9–7 pyramid filters, PKVA directional filtering, four multiscale levels, and six independent coordinate groups constructed in the valid P3+P4 range. The pool contains 245,760 eligible coordinates, providing a practical margin above the required 222,360-bit transport.

Before the profile entered the final experiment, it passed reconstruction, capacity, writability, self-gain, cross-talk, off-target-energy, and dense sign-recovery checks. The active toolbox, function resolution, bridge implementation, band inventory, and transform fingerprint are preserved with the evidence.

This gives the transport an explicit answer to a practical question that is often left implicit: not only which transform is used, but exactly which independently writable multiscale degrees of freedom carry the payload.

4.4 Capacity and realised-quality controls

The allocator selects exactly 222,360 unique coefficient slots and cannot exceed the capacity of any eligible group. A binary coefficient-map SHA-256 identifies the completed map.

Embedding follows

$$c' = c + \lambda w(2b - 1), \quad (2)$$

where c is the selected coefficient, b is the embedded bit, w is the local weight, and λ is the global strength.

A fixed-iteration binary search selects the largest feasible λ that remains inside the target quality interval after inverse transformation, clipping, half-up rounding, and uint8 conversion. All configurations therefore carry the same payload and meet the same realised cover–stego PSNR target of 45.0(1) dB. Any measured design advantage consequently comes from how protection and placement are organised, not from a smaller message or a larger visible distortion.

Supplementary Notes S1–S3 provide the complete header layout, seed derivation, allocation invariants, transform audit, and coefficient-map evidence.

5 Methods

5.1 A self-describing protected payload

The transport begins with a fixed protected header. The raw header contains 127 bytes and is encoded as one RS(255,127) codeword, producing 2,040 embedded header bits. It records the format and method identifiers, ECC layout, image dimensions, Base and Detail codeword counts, padding lengths, payload size, interleaver and scrambler identifiers, payload CRC32, and the SHA-256 identity of the canonical configuration.

The header occupies a fixed set of known coordinates and is embedded with unit weight. It is neither scrambled nor interleaved, allowing the receiver to recover the information required to interpret the remainder of the stream. This turns the hidden message into a versioned digital object whose construction can be checked rather than inferred.

Base and Detail then follow independent Reed–Solomon, deterministic scrambling, and deterministic interleaving paths. C0, C1, and C2 alternate complete Base and Detail transport blocks. C3 places all Base blocks before Detail so that the adaptively ranked coordinate map can give the perceptually dominant layer earlier access to preferred coordinate groups.

5.2 Validity-aware decoding

The receiver first extracts and validates the protected header. The expected format, method, dimensions, coding layout, configuration identity, and header CRC must agree before the body is interpreted.

A complete secret is accepted only after the required Reed–Solomon codewords and the payload CRC have been validated. This protects the scientific meaning of a successful decode: an output image is not presented as recovered merely because some extracted pixels look plausible.

An unsuccessful complete decode still contains useful information. The evaluation record therefore retains, where available, the stage, layer, codeword index, corrected-symbol counts, raw BER, Base and Detail BER, known-bit fraction, and CRC states. Partial correction remains observable as structured evidence even when the complete secret is not emitted.

5.3 Audited transform interpretation

The final profile uses 9–7 pyramid filters, PKVA directional filtering, four multiscale levels, and independent P3+P4 range coordinates. The adapter verifies that the required PDFB functions resolve inside the designated toolbox and records the toolbox identity, bridge, band inventory, coordinate construction, numerical environment, and transform fingerprint.

The profile entered the final experiment only after satisfying reconstruction, capacity, writability, self-gain, cross-talk, off-target-energy, and dense sign-recovery checks. This provides an explicit, reproducible transform interpretation and an independently writable coordinate map without assuming that every entry of a redundant directional representation is an independent degree of freedom.

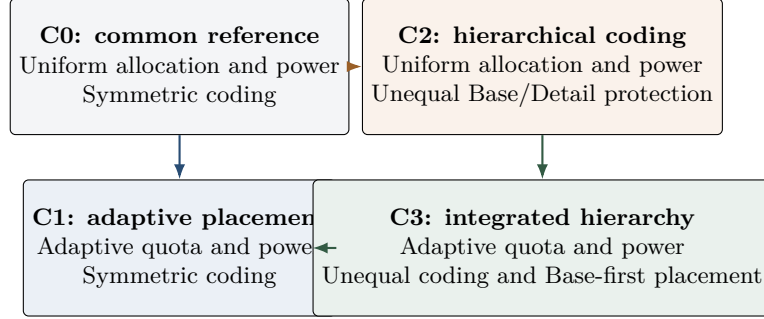


Figure 2: Mechanism-oriented C0–C3 comparison. C1 introduces adaptive placement, C2 introduces hierarchical coding, and C3 integrates adaptive quota, adaptive power, unequal protection, and Base-first placement.

5.4 Calibration-bound adaptive scoring

Adaptive scores are derived only from designated non-reporting calibration pairs. For each eligible coordinate group, the system combines energy, variance, entropy of absolute coefficient magnitudes, and stability under JPEG quality 70, Gaussian variance 10, and salt-and-pepper density 0.03.

The resulting stability artefact records the calibration inputs, attack definitions, feature values, transform fingerprint, and configuration identity. The final runner uses it only when its transform fingerprint matches the active transform. This binding keeps the meaning of the learned stability evidence connected to the representation on which it was measured.

5.5 Mechanism-oriented C0–C3 design

The experiment evaluates four configurations that progressively introduce the principal design mechanisms. Figure 2 summarises their relationship.

C0 provides the common reference with uniform allocation, unit embedding weights, and symmetric coding. C1 introduces adaptive quota and power while retaining symmetric coding. C2 redirects stronger Reed–Solomon protection toward Base while retaining uniform allocation. C3 integrates adaptive allocation, adaptive power, unequal coding, and Base-first placement.

All configurations carry the same payload, use the same transform profile, and satisfy the same realised-quality rule. The design therefore focuses the comparison on the organisation of protection and placement.

5.6 Fixed traceability cases and channel conditions

Four fixed cover–secret assignments produce sixteen saved embedding objects. Each saved stego image is reused across its channel evaluations so that embedding variation is not mixed with channel variation.

The core schedule contains clean transmission, JPEG quality 70, Gaussian variance 10, and salt-and-pepper density 0.03. It therefore produces 64 evaluation rows from the 16 embeddings. Predeclared triggers activate hard checks at JPEG quality 50, Gaussian variance 15, and salt-and-pepper density 0.05. Each activated family evaluates C0 and C3 on the four fixed pairs, and the full schedule is capped at 88 rows.

The pair, secret, payload, transform fingerprint, PSNR rule, and channel realisation remain aligned across methods. Recorded outcomes include cover–stego quality, raw BER, Base and Detail BER, header and payload validity, valid-secret quality when available, selected strength, runtime, memory, and immutable object identifiers.

The primary complete-validity measure is effective unrecovered bit rate, EUR: zero denotes valid complete recovery, while one denotes that the complete validity requirements were not met. Layer-specific and codeword-level records complement this strict endpoint by showing how information behaved before final validation stopped.

5.7 Content-addressed and interruption-resilient execution

Each embedding and channel evaluation is stored as an immutable content-addressed object. Its identity is derived from the inputs that determine the numerical result, including code, configuration, image content, calibration artefact, transform identity, and schema. An existing object is reused only after schema and SHA-256 validation.

The runtime path is tested under a real interruption. A worker process group is terminated with SIGKILL, after which the execution resumes. The test verifies that completed objects remain unchanged, unfinished work continues, incomplete attempts are not promoted to final results, inconsistent cache entries are quarantined, and the final archive is reconstructed and validated.

This execution model allows a long scientific run to continue from verified progress rather than depending on uninterrupted process lifetime or manually interpreted temporary files.

5.8 Scope of inference

The experiment is a focused four-case mechanism study. Results describe the fixed traceability pairs and channel realisations included in the protocol rather than an arbitrary image population. Extraction is semi-blind because the original cover supports coefficient differencing. Geometric attacks and steganalysis are outside the final C0–C3 matrix, and deterministic scrambling supports reproducibility and error dispersion rather than encryption.

These scope boundaries keep the relationship between the implemented evidence and the reported claims clear while preserving a strong foundation for broader follow-up studies.

5.9 Research ethics and AI-assisted editing

The study used digital image files and software experiments only. It involved no human participants, personal data, animal experiments, or biological materials; institutional ethics approval and informed consent were therefore not applicable.

OpenAI ChatGPT was used during manuscript development for language editing, structural revision, and LaTeX drafting. It was not used to generate experimental observations or replace the locked numerical pipeline. The author checked the calculations, references, code changes, figures, and final scientific claims and remains responsible for the manuscript.

6 Results

The experimental programme completed all sixteen embeddings and all eighty-eight scheduled evaluations. The archive contains 1,582 files, with the complete execution table reported in Supplementary Table S1.

The results tell a consistent three-part story. The transport worked in every clean case; C3 delivered the intended structural advantage for Base under the common payload and quality budget; and the selected attacked conditions lay beyond complete-payload recovery for every configuration.

6.1 Clean recovery confirmed the complete transport

All sixteen clean streams decoded successfully, giving $\text{EUR} = 0$ in every method–pair combination. This result confirms the whole path rather than only the embedding equation: layered payload construction, Reed–Solomon coding, deterministic coordinate selection, PDFB embedding, inverse transformation, uint8 quantisation, semi-blind extraction, and final integrity validation all remained mutually consistent.

The fixed payload fitted the independently writable coordinate pool, and sender and receiver reconstructed the same slot map. Exact recovery also survived the complete image-formation boundary at the common 45.0(1) dB cover–stego target. These clean cases establish the operating point from which channel effects can be interpreted.

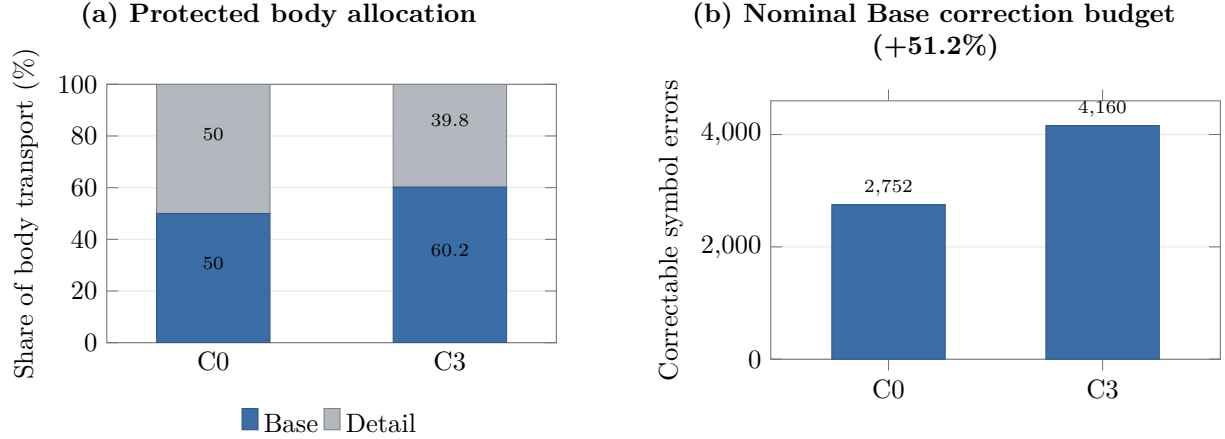


Figure 3: Base-layer design advantage. C3 reallocates the fixed transport toward Base and increases its nominal per-codeword-bounded correction budget. Full codeword counts and limits are listed in Supplementary Table S2.

6.2 C3 delivered the intended Base-layer advantage

Figure 3 presents the strongest measured design gain. C3 directs a larger share of the protected body toward Base and replaces every weaker Base codeword with RS(255,127). The nominal correction budget consequently rises from 2,752 to 4,160 symbols, a 51.2% increase.

Base grows from half of the protected body to 60.2%, and twenty-two blocks move from a 32-symbol to a 64-symbol correction radius. This advantage follows from the implemented transport structure; it does not depend on choosing a favourable image or channel realisation.

Placement reinforces the coding decision. Once adaptive coordinate groups are ranked, C3 writes the complete Base stream before Detail. The perceptually dominant layer therefore receives stronger codewords and earlier access to the coordinates preferred by calibration.

6.3 The comparison retained a common operating budget

Each configuration carried the same 222,360 protected bits. Its global embedding strength was selected independently, but the realised stego image had to remain within the shared PSNR interval after inverse transformation, clipping, rounding, and uint8 conversion.

The comparison is therefore controlled in two practical ways: no method hides less information, and no method is allowed more visible distortion. Differences arise from the organisation of coding and placement. For C1 and C3, adaptive scores determine capacity-safe integer quotas and bounded local weights; the resulting coefficient maps are deterministic and identified by SHA-256 digests.

6.4 Representative visual evidence

Figure 4 presents the first pair in the locked manifest, selected before visual inspection. The panel is generated directly from archived image objects and brings together the cover, secret, C0 and C3 stego images, amplified absolute-difference maps, and the clean recovered secret. Cover-stego PSNR and clean-recovery status are computed by the same deterministic generator and printed on the figure.

6.5 The attacked conditions located the recovery boundary

Every tested JPEG, Gaussian, and salt-and-pepper stream produced $EUR = 1$, and the same family-level pattern continued in the predeclared hard checks. Figure 5 summarises the operating map: clean recovery was universal, whereas the selected attacked conditions remained outside the region of complete-payload validity.

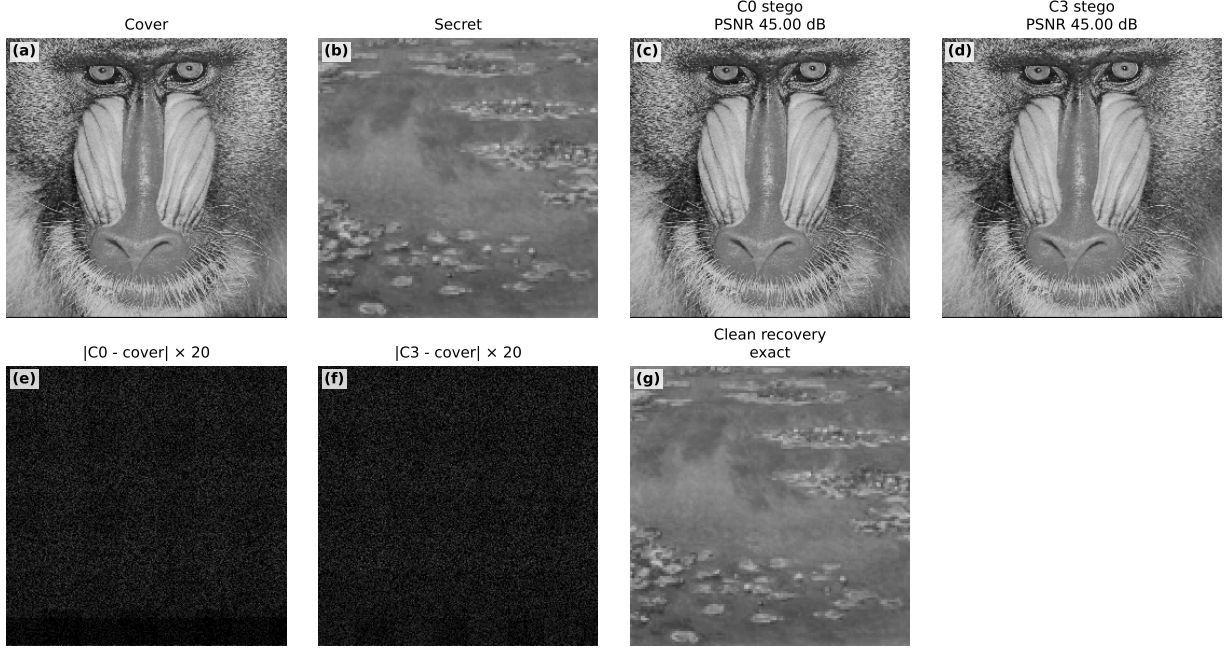


Figure 4: Pre-specified representative image panel. Stego images and amplified absolute-difference maps are shown together so that visual similarity and embedding structure can be inspected in the same view. Source and output SHA-256 identities are recorded in `figures/representative-pair.json`.

This outcome separates design margin from end-to-end validity. C3 gives Base more protection, but complete recovery still depends on the header and every required payload component. One essential codeword beyond its correction radius can therefore invalidate a stream that still contains many corrected blocks.

Because all clean cases succeeded, the attacked boundary cannot be attributed to insufficient capacity, mismatched coordinate maps, or a general transform failure. The archived codeword outcomes, corrected-symbol counts, layer errors, CRC states, and failure stages preserve the information needed to examine how each configuration approached that boundary.

7 Discussion

The clearest insight from this study is the difference between strengthening a layer and recovering an entire payload. C3 gave Base more redundancy and earlier access to the highest-ranked coordinate groups. That improvement is real and measurable. At the same time, the selected attacked conditions remained beyond the point at which the additional margin was sufficient for complete-payload validation.

Seeing both results together is more informative than reducing the experiment to a single pass-fail statement. The architecture reveals what became stronger; the channel results show where that strength was no longer enough. In this operating regime, hierarchical protection increased the design margin without yet moving the complete-recovery boundary.

The fixed quality target helps explain why. Coefficient displacement is deliberately limited, while the protected stream occupies a broad coordinate pool. Recovery also depends on a valid header and every required payload component. A stream can therefore retain substantial correct information and still be rejected when one essential codeword crosses its correction radius.

This strict decision rule is appropriate for claiming exact recovery, but it may understate the practical value of unequal protection. Base was designed to preserve the visually dominant structure of the secret. Its advantage may become visible first as a valid lower-fidelity reconstruction rather than as immediate recovery of both layers. A separate Base integrity field would make that intermediate outcome measurable without weakening the existing complete-payload criterion.

	Clean	JPEG	Gaussian	S&P
C0	✓	×	×	×
C1	✓	×	×	×
C2	✓	×	×	×
C3	✓	×	×	×

Figure 5: Complete-payload validity by method and channel family. A check denotes valid recovery for all clean cases; a cross marks channel conditions beyond complete recovery for every evaluated case in that family.

The completed run is well suited to such follow-up analysis because its history is preserved. Each result remains connected to a fixed pair, method, transform fingerprint, coefficient map, embedding object, attack realisation, and evaluation object. New severity levels or layer-specific validity rules can extend the evidence without rewriting the operating point reported here.

Equally important, the comparison was not adjusted after the outcome was known. The payload stayed fixed, the PSNR target was not relaxed, image pairs were not exchanged, and extra channel settings were not selected to manufacture a favourable result. That restraint gives the boundary its meaning.

Taken together, the evidence supports three conclusions. The digital transport is operational, as shown by exact clean recovery in every case. The hierarchical design genuinely strengthens Base under the fixed budget. And the tested channels define a reproducible region beyond complete recovery, providing a clear reference for finer threshold mapping and layer-aware validation.

8 Limitations and Future Work

The evidence is intentionally bounded to four fixed traceability cases. The PDFB profile is an explicit independent implementation rather than an author-equivalent reconstruction, and extraction remains semi-blind. Geometric robustness and steganalysis are outside the final factorial experiment.

These limits point to focused extensions rather than a redesign of the completed algorithm. A finer severity sweep between clean recovery and the present medium settings would show where header validity, codeword margins, Base and Detail errors, and complete-payload validity begin to diverge. The archived diagnostics already retain the information needed to support that analysis.

Layer-specific integrity is the most direct next step. It would distinguish complete recovery, valid Base recovery without Detail, and insufficient integrity for any output. Such a rule would align the decoding decision with the semantic hierarchy already present in the payload and would reveal whether stronger Base protection preserves a meaningful approximation before the full secret becomes recoverable.

The same fixed transform interpretation can also support comparisons across payload rates, metadata protection choices, and Base/Detail redundancy allocations. These would be registered as new experiments while leaving the present evidence unchanged. Direct numerical comparison with the source article should wait until transform, payload, pairing, quantisation, extraction, attacks, and metrics are fully harmonised.

9 Conclusion

This work turns contourlet embedding into a layered digital transport whose priorities are explicit. Information that contributes most strongly to the secret image is separated as Base, given stronger Reed–Solomon protection, and placed earlier in adaptively ranked multiscale coordinates.

Under the fixed payload and visual-quality budget, C3 raises the nominal Base correction capacity by 51.2% and expands its share of the protected body to 60.2%. Every clean stream completed the full encoding, embedding, extraction, and integrity-validation path. The selected JPEG, Gaussian, and salt-and-pepper conditions then marked a reproducible boundary beyond complete-payload recovery.

The contribution is therefore both architectural and empirical. Semantic layering, transform-aware placement, independently writable PDFB coordinates, self-describing validity, and traceable execution are combined in one inspectable system. The result is not a claim of universal robustness; it is a verified account of what was strengthened, where the current operating point ends, and how the evidence can be extended without losing its history.

Data Availability

The repository branch `agent/runtime-resume-gate` contains the executable reconstruction, configurations, manuscript sources, audit documents, the published reproduction capsule, and the validated result archive. Exact filenames and SHA-256 identities are reported in Supplementary Information. Reuse of image-derived material remains subject to the provenance and rights notices stored with the archived inputs.

Code Availability

The numerical execution is linked to commit `7ff0c5ab` and run `f7acf6d9d31dd66cddf1`; full immutable identifiers are reported in Supplementary Information and the repository audit records. The manuscript, deterministic figure generators, and provenance documentation are maintained on `agent/runtime-resume-gate`. The submitted source tree should receive a release tag or archived DOI so that the reporting snapshot remains immutable.

Author Contributions

A.A. contributed to conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, visualisation, writing of the original draft, review and editing, and project administration. This CRediT-style statement must be confirmed by the author before submission.

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Competing Interests

The author declares no competing interests.

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